

Parallel and distributed Computing

Lecture 22

Bazam-e-Jahan

Lecturer

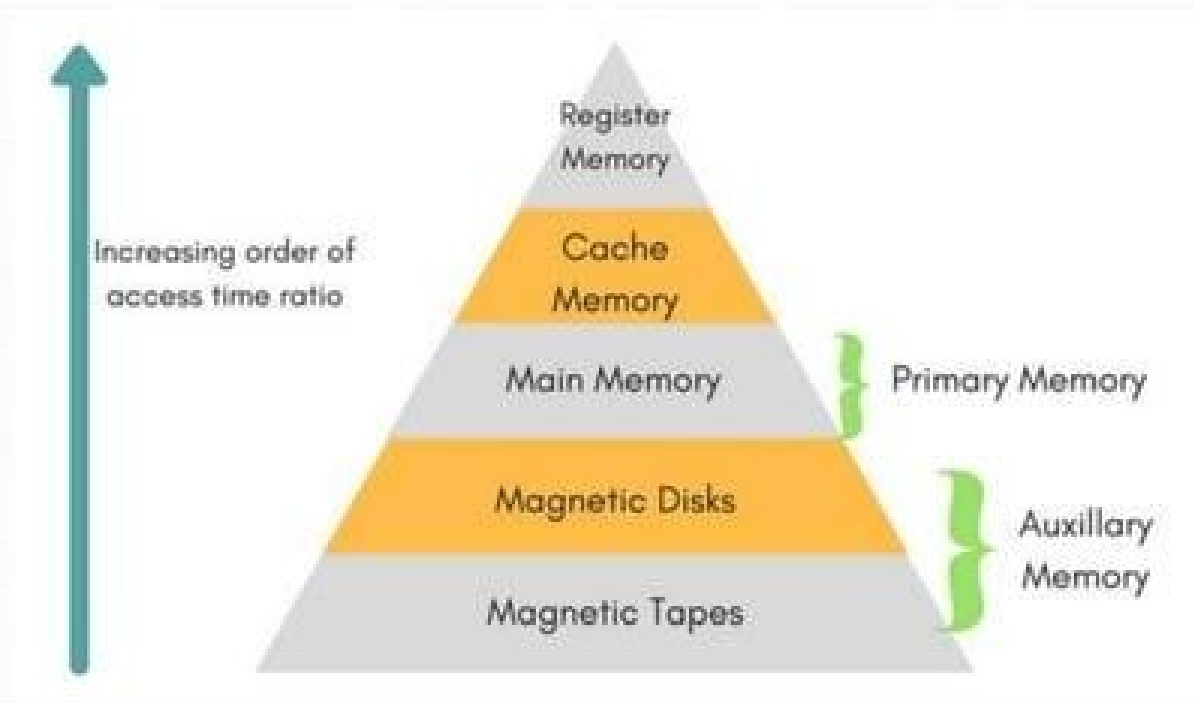
The University of Faisalabad

Memory hierarchies

Memory Hierarchy

- Computer Memory Hierarchy is a pyramid structure. A memory is just like a human brain. It is used to store data and instructions
- The memory unit is an essential component in any digital computer since it is needed for storing programs and data.
- The memory unit that directly communicate with CPU is called the main memory .
- Devices that provide backup storage are called auxiliary memory.
- A special very-high-speed memory called cache is used to increase the speed of processing.

Diagram of Memory Hierarchy



Auxiliary Memory

- Auxiliary memory is also called **secondary memory**.
- Auxiliary memory Is a device which provide backup storage .
- It is slower than the main memory
- The contents of secondary memories are first transferred to the main memory, and then the CPU can access it.
- For Example:- Magnetic tap , 1 disk, cd-rom etc.



Magnetic Tape

- Used for backup, and storage of less frequently used information.
- If the tape is damaged, the data is lost.
- Typically stores from 20 GB to 200 GB.
- Magnetic tapes are less expensive.
- Magnetic tape allows fast sequential accessing but is slower in random accessing.



Magnetic Tap

- magnetic disk is used as a **secondary storage** in modern computers.
- The magnetic disk is fast in accessing data sequentially or randomly.
- Magnetic disk can be updated.
- storage capacity of the magnetic disk is from several hundred GB to Tera bytes.
- Magnetic disk is more expensive.



Primary Memory (Main Memory)

- Primary memory holds only those data and instructions on which the computer is currently working.
- It has a limited capacity and data is lost when power is switched off.
- The data and instruction required to be processed resides in the main memory.
- It is the working memory of the compute.
- A computer cannot run without the primary memory



RAM

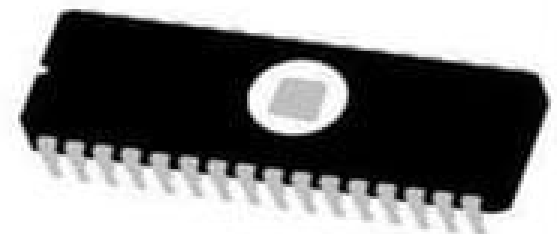
- RAM stands for **Random Access Memory**.
- RAM is the internal memory of the CPU for storing data, program, and program result.
- It is a read/write memory which stores data until the machine is working.
- RAM can be accessed randomly but it is very expensive.
- RAM is volatile, i.e. data stored in it is lost when we switch off the computer.
- Types of RAM - DRAM, SRAM.



RAM

ROM

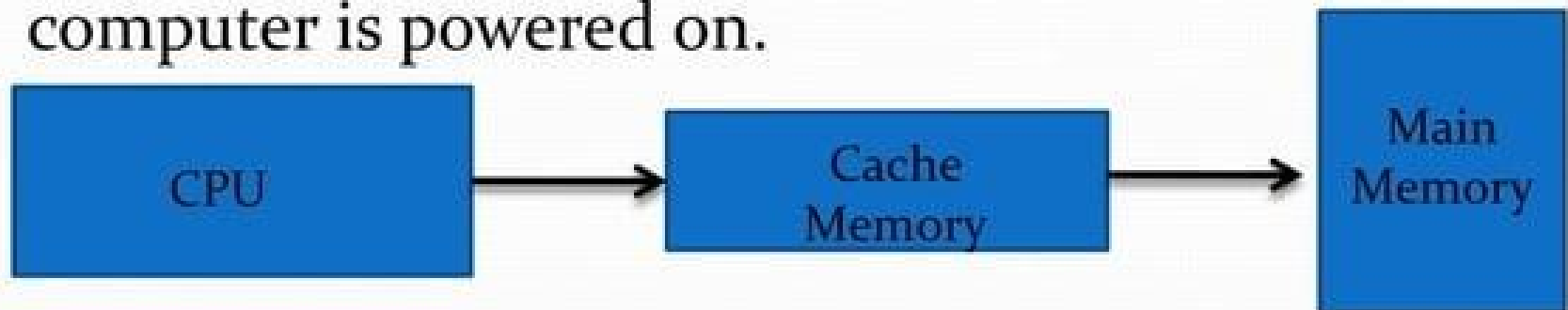
- ROM stands for **Read Only Memory**.
- we can only read but cannot write on it.
- It is non-volatile memory.
- Data in ROM can not be modified.
- ROM is comparatively smaller than RAM
- Types of ROM are PROM, EPROM, EEPROM.



ROM
Read only Memory

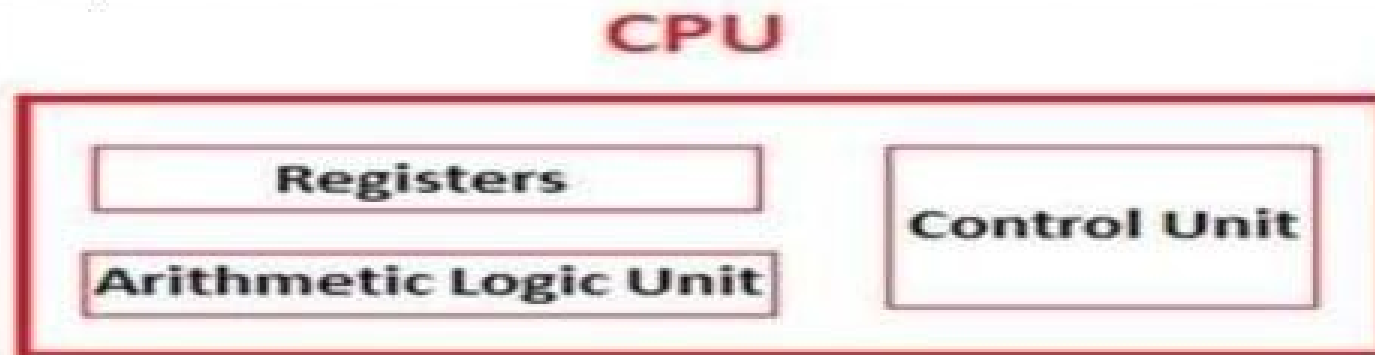
Cache Memory

- Cache memory is also called Temporary Memory.
- Cache memory is in small size, type of volatile memory that provides high speed data access to a processor.
- It stores frequently used computer programs, applications and data.
- It stores and retrieves the data only until a computer is powered on.



Register Memory

- Register memory is the smallest and fastest memory in a computer.
- It is located in the CPU in the form of registers.
- A register temporarily holds frequently used data, instructions and memory address that can be quickly accessed by the CPU.
- Types of Register Memory Address Register, Memory Buffer Register, Memory Data Register (MDR)



Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.